

SUMMARY TABLE: MATHEMATICS GRADE 10

Sub-Strand 1.4: Congruence — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 1.4: Congruence
Driving Question	DRIVING QUESTION: "How does the mathematics of reflection explain why the mirror image of a boat on Lake Victoria is always a perfect, same-sized copy — and how can we use that same idea to prove that two shapes are truly identical?"

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Finding Balance: Lines of Symmetry	Students observed that the Lake Victoria boat reflection appears exactly the same size as the real boat, that the waterline divides the image into two matching halves, and that folded cut-out shapes (fish, boat, acacia leaf) have fold lines along which the two halves align exactly.	Students learned that a line of symmetry is a line that divides a figure into two congruent halves that are mirror images of each other, that the waterline in the Lake Victoria photograph acts as a line of symmetry, and that the word CONGRUENT describes two shapes that are exactly the same in shape and size.	Students began to explain that the reason the reflected boat image is the same size as the original is connected to the way the waterline acts as a line of symmetry — but the full mathematical explanation (reflection as a rigid transformation preserving all lengths and angles) remains open for investigation across the remaining lessons.
Lesson 2 Flipping Across the Axes: Reflection on the Cartesian Plane (x- and y-axes)	Students watched the ARES videos on reflecting points and objects on the Cartesian plane, folded a printed grid along the x-axis and y-axis to physically locate reflection images of a boat polygon, and measured the perpendicular distances from each vertex to the line of reflection.	Reflecting a point across the x-axis maps $(x, y) \rightarrow (x, -y)$; reflecting across the y-axis maps $(x, y) \rightarrow (-x, y)$. In both cases, each image point is exactly the same perpendicular distance from the line of reflection as the original point. The image is congruent to the original figure.	The waterline in the Lake Victoria phenomenon acts as the x-axis: the water surface applies the rule $(x, y) \rightarrow (x, -y)$ to every point of the boat simultaneously, which is why the reflection is always the same size and always appears the same distance below the waterline as the boat sits above it.
Lesson 3 Shifting the Mirror: Reflection Along Lines Parallel to the Axes	Using the matatu district map on the Cartesian plane, students plotted landmark points and counted perpendicular grid squares to find their images across $y = 2$, $y = \text{minus}4$, $x = 3$, and $x = \text{minus}1$, recording original and image coordinates in a table.	Reflecting a point across $y = k$ maps (x, y) to $(x, 2k \text{ minus } y)$; reflecting across $x = k$ maps (x, y) to $(2k \text{ minus } x, y)$. In both cases the image is the same perpendicular distance from the mirror line as the original, and the reflection for $k = 0$ is the special case discovered in Lesson 2.	If the waterline of Lake Victoria were at any height k above the standard axis — for example after seasonal flooding — the boat reflection would still land at an equal perpendicular distance below that new waterline, because the equal-distance rule works for any horizontal mirror line, not just $y = 0$.

Lesson 4 Diagonal Mirrors: Reflection Along $y = x$ and $y = -x$	Students observed that when triangle ABC was reflected across the line $y = x$ on a Cartesian grid, each vertex landed at a point where the x- and y-coordinates were swapped — for example, $A(1, 3)$ mapped to $A'(3, 1)$. They also observed that when reflecting across $y = -x$, both coordinates were swapped and negated — for example, $P(2, -1)$ mapped to $P'(1, -2)$. In both cases, students measured and confirmed that each vertex was the same perpendicular distance from the mirror line as its corresponding image vertex.	Students learned that: (1) reflecting a point across $y = x$ follows the rule $(x, y) \rightarrow (y, x)$; (2) reflecting a point across $y = -x$ follows the rule $(x, y) \rightarrow (-y, -x)$; (3) the mirror line in any reflection is always equidistant from object and image (each point and its image are the same perpendicular distance from the line of reflection); (4) the image produced by reflection is always congruent to the original figure — same shape, same size; (5) diagonal reflections are not the same as rotations, even though they can look similar.	Students explained their observations using the Claim–Evidence–Reasoning framework: they claimed the coordinate rules for $y = x$ and $y = -x$, cited their measured coordinate data as evidence, and reasoned that the rules work because the perpendicular-distance property of reflection forces an exact coordinate swap (for $y = x$) or a negate-and-swap (for $y = -x$). They connected this back to the Lake Victoria phenomenon by noting that no matter what orientation the mirror line takes — horizontal, vertical, or diagonal — the reflected boat image is always congruent to the original, because the fundamental property of reflection (equidistance from the mirror line) never changes.
Lesson 5 Reverse Engineering the Mirror: Finding the Equation of the Mirror Line	Students observed (via video and hands-on grid activity) that the perpendicular bisectors of the segments joining each object point to its corresponding image point all intersect at — or lie along — a single straight line. This line is the mirror line. Students measured and confirmed that each object point and its image are equidistant from this line.	Students learned that the line of reflection can be determined from an object-image pair alone, without being told where the mirror is. The mirror line is the perpendicular bisector of the segment joining any object point to its image. Its equation can be written in the form $y = mx + c$, where m is found from the gradient of the line through two midpoints and c is found by substitution.	Students explained how the equal-perpendicular-distance property of reflection makes it possible to 'reverse-engineer' the hidden waterline of Lake Victoria: by joining the boat's key points to their mirror images, finding the midpoints of those segments, and constructing perpendicular bisectors, the exact position and equation of the water surface can be determined mathematically — even when it is obscured by mist.
Lesson 6 Same Shape, Same Size: Introducing Congruence	Students observed (via the anchor photograph, the congruence video, and the card-triangle sorting investigation) that triangles which match on all three sides — as occurs between a reflected image and its original across the Lake Victoria waterline — are always identical in shape and size, regardless of how the triangles are oriented.	Students learned that reflection always produces an image that is congruent to the original figure, because every point and its image are the same perpendicular distance from the line of reflection, preserving all side lengths. They learned the five formal congruence conditions (SSS, SAS, ASA, AAS, RHS) that are sufficient to prove two triangles identical, and that SSS alone is enough to prove the reflected boat image congruent to the original boat.	Students explained that the reflected boat image on Lake Victoria is a perfect, same-sized copy of the original because reflection is a distance-preserving transformation: each vertex of the boat maps to a vertex in the image at an equal perpendicular distance from the waterline, so all three side lengths of the image triangle equal all three side lengths of the original triangle — satisfying SSS — which proves the two triangles are congruent (Triangle PQR $\cong$ Triangle P'Q'R').
Lesson 7 Proving It: Applying Congruence Conditions	We observed three pairs of triangles and a reflected fishing-boat coordinate diagram, and we saw that the side lengths of the original triangle and its reflected image are always equal.	We learned how to write a formal congruence proof by naming the correct condition (SSS, SAS, ASA, AAS, or RHS), listing all pairs of equal elements with their values, and writing the congruence statement with vertices in correct	We explained that the Lake Victoria boat reflection is always a perfect, same-sized copy because reflection is a rigid transformation that preserves all side lengths — so SSS is always satisfied, and the reflected image is always congruent to

		correspondence order.	the original boat.
Lesson 8 Mathematics in the Real World: Applications of Reflection and Congruence	At dawn on the shore of Lake Victoria near Kisumu, a fisherman's boat casts a perfect mirror image on the still water. The reflected image appears identical in shape and size to the real boat — yet it is flipped. No matter how complex the boat's shape, the reflection is always the same size and always appears at an equal distance on the other side of the waterline. We also observed: a Maasai beadwork necklace where corresponding beads are equidistant from a central line of symmetry; a surveyor at the Rift Valley using congruent triangles to measure an inaccessible gorge width; and an architect's building blueprint where the two wings are mirror images of each other.	We learned that: (1) Reflection maps every point in a figure to a corresponding point on the opposite side of the line of reflection, at exactly the same perpendicular distance — this is the rule the water surface follows. (2) Any non-vertical line of reflection can be expressed in the form $y = mx + c$; a vertical one is written $x = k$. (3) The image produced by reflection is always congruent to the original figure — same shape, same size — because reflection is an isometry (a distance-preserving transformation). (4) Congruent triangles can be used to calculate distances that cannot be measured directly. (5) These properties of reflection and congruence appear in Maasai design, land surveying, architecture, and the natural world.	The perfect mirror image of the fisherman's boat at Lake Victoria is always the same size as the real boat because reflection is an isometry — it preserves all distances. The waterline acts as the line of reflection (expressible as $y = mx + c$). Every point on the boat maps to a corresponding image point that is exactly the same perpendicular distance from the waterline, but on the opposite side. Because every distance is preserved, the image is congruent to the original boat — identical in shape and size, but flipped. This is why the reflection never stretches, shrinks, or distorts the boat's image, no matter how far the boat is from the water's surface.

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